Diffusion-guided Intrinsic Curiosity Module for Sparse environments.

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Abstract

Learning effective control policies in long-horizon, sparse-reward environments remains a central challenge in reinforcement learning, where agents often fail to discover successful behaviors due to limited feedback and poor exploration. In this work, we present a hybrid exploration–planning framework that combines Intrinsic Curiosity Module (ICM)–driven exploration, small sets of human expert demonstrations, and a goal-conditioned diffusion model for trajectory generation. First, an ICM-augmented policy is trained to overcome sparse rewards and produce broad, diverse coverage of the state space. We then combine these exploratory transitions with limited expert demonstrations to train a diffusion model that captures the manifold of feasible, structured, and goal-directed trajectories. This model functions as a long-horizon planner: given a start and goal state, it generates smooth multi-step trajectories that correct for the myopic and value-local nature of DQN in sparse domains. On the challenging FrozenLake-8×8 environment, our method substantially improves success rates and planning reliability compared to curiosity-augmented DQN alone. These results demonstrate that integrating broad exploration, minimal expert guidance, and diffusion-based trajectory modeling provides an effective, data-efficient solution for decision-making in long-horizon sparse-reward tasks.

1 Introduction

Reinforcement learning (RL) agents frequently struggle in environments where rewards are sparse, delayed, or dependent on achieving long-horizon goals. Traditional value-based methods such as Deep Q-Networks (DQN) have shown impressive performance in dense-reward, short-horizon tasks but tend to collapse when exploration becomes essential. In sparse settings, the agent seldom encounters rewarding states through random exploration, causing both the replay buffer and Q-value targets to be dominated by uninformative transitions. As a result, training becomes unstable, inefficient, and often fails entirely.

One family of methods designed to address insufficient exploration are curiosity-driven intrinsic motivation models, particularly the Intrinsic Curiosity Module (ICM). ICM introduces an auxiliary reward signal derived from prediction error: if the agent cannot accurately predict the forward dynamics (i.e., next state given the current state and action), the resulting error is treated as intrinsic reward. This encourages the agent to visit novel or uncertain states. While ICM improves exploration significantly, its impact is fundamentally local—it drives the agent toward surprising states, not necessarily goal-relevant states. In long-horizon problems where the optimal sequence of states forms a narrow path through the state space, curiosity alone is insufficient: the model may explore broadly without discovering the reward-relevant trajectory. Another orthogonal limitation is that curiosity and DQN remain reactive rather than planning-based: they do not explicitly model global trajectories or future state distributions. Once the environment requires coherent sequences of transitions extending far into the future, purely local, step-based learning often fails. Recent generative modeling advances

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offer a complementary direction. Diffusion models, originally developed for image synthesis, have been extended to sequence modeling, planning, and control. When conditioned on start and goal states, diffusion models can learn to generate feasible, long-horizon transition sequences that connect these endpoints. Instead of predicting single actions or Q-values, diffusion models learn the distribution of entire trajectories. When combined with RL, such models can supply the agent with candidate plans, augment replay buffers, and help bootstrap learning in environments where direct exploration is insufficient. In this project, we investigate how curiosity-driven exploration (via ICM) and goal-conditioned diffusion trajectory modeling can be combined to overcome sparse reward limitations in DQN-based agents. The major key ideas are:

1. ICM ensures that the RL algorithm explores all possible transitions, even in the absence of rewards.
2. Human-provided expert transitions give the diffusion model an anchor for desirable long-horizon behavior.
3. Goal-conditioned diffusion models learn to generate full state sequences that connect arbitrary start and end states.

In environments where ICM alone is sufficient, eg simple, short-horizon tasks, such as FrozenLake 4x4, diffusion modeling provides no additional benefit. However, in more challenging settings with long-horizon dependencies, such as FrozenLake 8x8, diffusion modeling becomes crucial. It allows the agent to model and sample structured sequences that rarely occur through random or curiosity-based exploration. The resulting hybrid architecture explores more intelligently, learns more robustly, and can generate coherent long-horizon plans even with minimal external reward.

2 Prior Work

2.1 Intrinsic Motivation and Curiosity-Driven Exploration

Intrinsic Curiosity Modules (ICM) [1] and related methods (e.g., Random Network Distillation [2], Episodic Curiosity [3]) introduce self-generated reward signals that encourage agents to explore novel or unpredictable states. ICM, in particular, learns a forward dynamics model and defines intrinsic reward as the prediction error of this model:

$$r_t^{\text{int}} = \|f_\phi(s_t, a_t) - s_{t+1}\|^2.$$

Such mechanisms have been shown to improve exploration in sparse-reward environments like Montezuma's Revenge and grid-based navigation tasks. However, intrinsic curiosity is local in nature, it encourages novelty without guaranteeing goal-directed behavior. This limitation is pronounced in long-horizon tasks, where discovering multi-step reward sequences is unlikely through exploration alone.

Other approaches combine intrinsic motivation with hierarchical or model-based methods to improve long-term planning. For example, Hierarchical RL decomposes tasks into subgoals [4], and count-based exploration uses state visitation statistics to encourage coverage [5]. While effective in some settings, these approaches can require significant tuning or additional model structure.

2.2 Generative Models for Trajectory Planning

Recent work has leveraged generative sequence models to address the challenges of long-horizon planning and sparse reward:

- Variational autoencoders (VAEs) have been used to learn latent representations of trajectories for planning and policy learning [6]
- Diffusion models, originally developed for image synthesis [7], have recently been adapted for trajectory generation in reinforcement learning [8; 9]. By modeling the distribution over sequences, these models can generate coherent multi-step trajectories conditioned on start and goal states, capturing long-range dependencies that standard value-based or policy-based RL cannot easily learn.

Diffusion-based trajectory modeling has been applied to robotic control [10], motion planning [9], and multi-agent trajectory prediction. Key advantages include the ability to:

- Learn complex, multimodal distributions over feasible trajectories
- Generate long-horizon sequences even when direct experience of full paths is limited
- Integrate heterogeneous data, including **exploratory trajectories** and sparse expert supervision

2.3 Combining Exploration and Generative Planning

A growing body of work explores hybrid approaches that combine exploration-driven data collection with generative planning models:

- Exploration-first strategies collect diverse transitions via intrinsic motivation or random policies and then use learned models to produce goal-directed sequences [11; 12].
- Model-based RL with learned trajectory priors uses generative models to propose multi-step actions or state sequences, effectively turning noisy exploratory experience into structured plans [8].

Our Approach: Our method builds on this idea by using ICM-driven exploration to collect diverse state transitions, augmenting them with expert transitions near the goal, and training a goal-conditioned diffusion model to generate coherent long-horizon trajectories. Unlike methods that rely on full expert demonstrations, hierarchical subgoals, or dense reward shaping, our method is data-efficient, requires minimal supervision, and scales naturally to sparse, long-horizon environments.

3 Background

3.1 Deep Q-Networks (DQN)

Deep Q-Networks (DQN) are a foundational value-based reinforcement learning algorithm for discrete action spaces. They learn an approximation $Q_\theta(s, a)$ of the state-action value function:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \;\middle|\; s_0 = s, a_0 = a \right],$$

where γ is the discount factor and r_t is the reward at step t. The policy is derived greedily:

$$\pi(s) = \arg\max_a Q_\theta(s, a).$$

DQN uses experience replay and target networks to stabilize learning. Transitions (s_t, a_t, r_t, s_{t+1}) are stored in a buffer, and minibatches are sampled to minimize the temporal-difference loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q_{\theta^-}(s', a') - Q_\theta(s, a) \right)^2 \right],$$

where θ^- are the parameters of a slowly updated target network. While effective in many dense-reward tasks, DQN struggles in sparse-reward environments, where the agent rarely experiences transitions leading to reward, causing Q-value updates to carry almost no informative signal.

3.2 Intrinsic Curiosity Module (ICM)

The Intrinsic Curiosity Module (ICM) addresses sparse reward challenges by generating internal rewards to encourage exploration. The core idea is to reward the agent for transitions that are difficult to predict, creating a surrogate reward that drives novelty-seeking behavior. ICM consists of two learned components:

1. Forward model $\phi(s_t, a_t)$, learns to predict the next state.
2. Inverse model $\psi(s_t, s_{t+1})$, learns to predict the action that causes transition from state s_t to state s_{t+1}.

The intrinsic reward is derived from the forward model prediction error:

$$r_t^{int} = \frac{\eta}{2}\|\hat{\phi}(s_{t+1}) - \phi(s_{t+1})\|_2^2$$

where η scales the intrinsic reward. The inverse model ensures the state representation captures information relevant to controllable aspects of the environment. This intrinsic reward is combined with the external reward:

$$r_t^{total} \;=\; r_t^{ext} \;+\; r_t^{int}$$

By training the agent with these augmented rewards, ICM encourages it to explore unfamiliar states, increasing coverage in sparse-reward domains. However, ICM alone does not guarantee that exploration is goal-directed. The agent may wander in novel regions without discovering the optimal trajectory.

3.3 Goal-Conditioned Diffusion Models

Diffusion models are probabilistic generative models originally developed for image synthesis. Recent work has extended them to sequence modeling and trajectory generation. In the RL context, diffusion models can learn a distribution over multi-step state sequences, making them particularly useful for long-horizon planning.

Forward and Reverse Processes

A diffusion model consists of:

1. **Forward (noising) process:** Gradually adds Gaussian noise to a sequence x_0 over T timesteps:

   $$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I),$$

   where β_t is a small variance schedule.
2. **Reverse (denoising) process:** Learns to remove noise iteratively:

   $$p_\theta(x_{t-1} \mid x_t) \approx q(x_{t-1} \mid x_t, x_0),$$

   allowing the model to sample coherent sequences starting from pure noise.

Goal Conditioning

For trajectory planning, diffusion can be conditioned on start and goal states:

$$p_\theta(\tau \mid s_{\text{start}}, s_{\text{goal}}),$$

where $\tau = (s_1, s_2, \ldots, s_T)$ is a sequence of states. The model can be trained on a mixture of:

- Curiosity-driven exploratory trajectories produced by the ICM-augmented DQN policy, and
- Expert transitions near the goal, which provide local guidance for reaching reward states.

During training, these trajectories are constructed from arbitrary start and end states rather than full task demonstrations. This allows the model to learn the underlying transition manifold and capture multi-step dependencies between states.

At evaluation, the trained diffusion model can be conditioned on a given start and goal state to generate a full trajectory connecting them. Sampling from the model produces coherent, goal-directed sequences, even when the agent has never experienced the full start-to-goal path in a single rollout. These generated trajectories can then be used to provide candidate long-horizon plans for direct execution by the agent. This approach converts broad, noisy exploration into structured long-horizon behavior, without requiring full expert demonstrations or reward shaping.

4 Methodology

Our approach leverages the Intrinsic Curiosity Module (ICM) to drive exploration in sparse-reward, long-horizon tasks, and then uses goal-conditioned diffusion models to convert exploratory transitions into structured trajectories. The methodology consists of three key steps: ICM-driven transition collection, diffusion model training, and trajectory sampling for planning.

4.1 Training the DQN model with Intrinsic Curiosity Module

The Intrinsic Curiosity Module (ICM) is built alongside a DQN agent to drive exploration:

1. Inverse model: predicts the action taken given consecutive states.
2. Forward model: predicts the next state embedding given the current state and action.

At each step, the forward model error is used to compute the intrinsic reward which is added to the extrinsic reward for DQN updates. The inverse and forward models are trained jointly with the agent, ensuring that the learned state embeddings encode controllable aspects and that novel transitions are properly incentivized. The network architecture is shown in Figure 1.

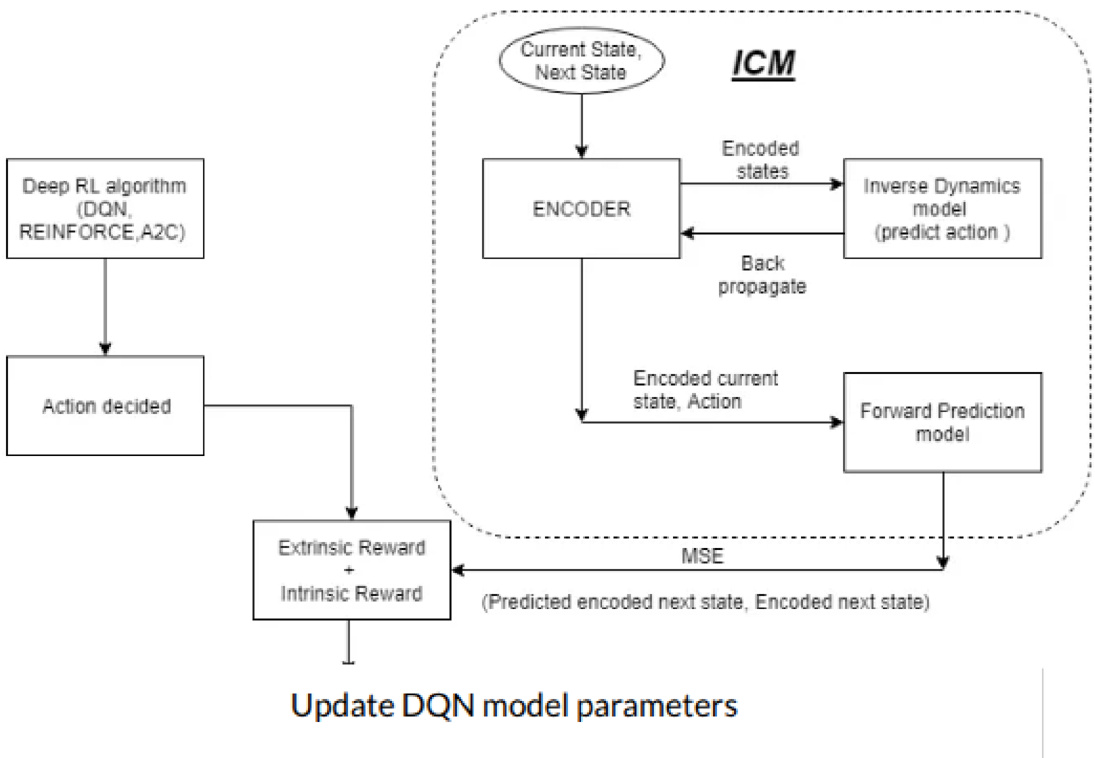

Figure 1: ICM + DQN model architecture.

4.2 Transition Collection

After training, we collect two types of data, ICM-driven exploratory transitions that cover a wide range of the state space, and near-goal expert transitions, which provide guidance in the vicinity of reward states. The combination of exploratory and near-goal transitions allows the model to learn both feasible dynamics and reward-relevant structure.

4.3 Diffusion Model Training

The diffusion model is trained on these windows to learn the distribution of multi-step trajectories.

1. Trajectory windows are sampled from arbitrary start and end states, ensuring coverage across the state space.
2. Trajectories are scored to ensure optimality.
3. The model is trained to denoise trajectories conditioned on start and goal states, learning to generate coherent sequences that connect arbitrary start and goal points.

4.4 Trajectory Sampling and Integration

At evaluation, the trained diffusion model can generate full trajectories given a start and goal:

- Initialize a noisy trajectory.
- Iteratively denoise conditioned on start and goal.
- Sample multiple candidates and score these.
- Execute the trajectory.

5 Results

5.1 DQN vs ICM

1. We observe that the ICM module consistently achieves maximum returns that the environment offers for both FrozenLake4x4 and FrozenLake8x8 environments. Figures 2 and 3 show moving average returns comparison for FrozenLake4x4 and FrozenLake8x8 respectively.
2. For short-horizon tasks, we observe that ICM is sufficient to reach goals reliably. However, even ICM often fails to reach the goal state for longer horizons.

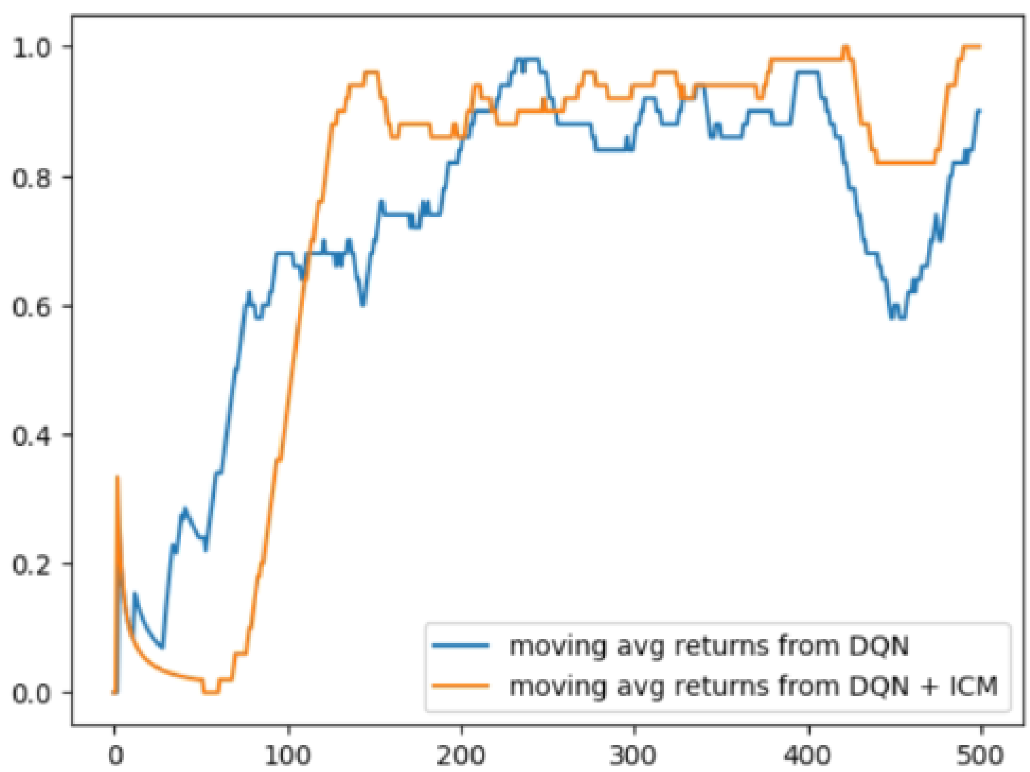

Figure 2: Moving average returns of standalone DQN vs DQN with ICM: FrozenLake4x4.

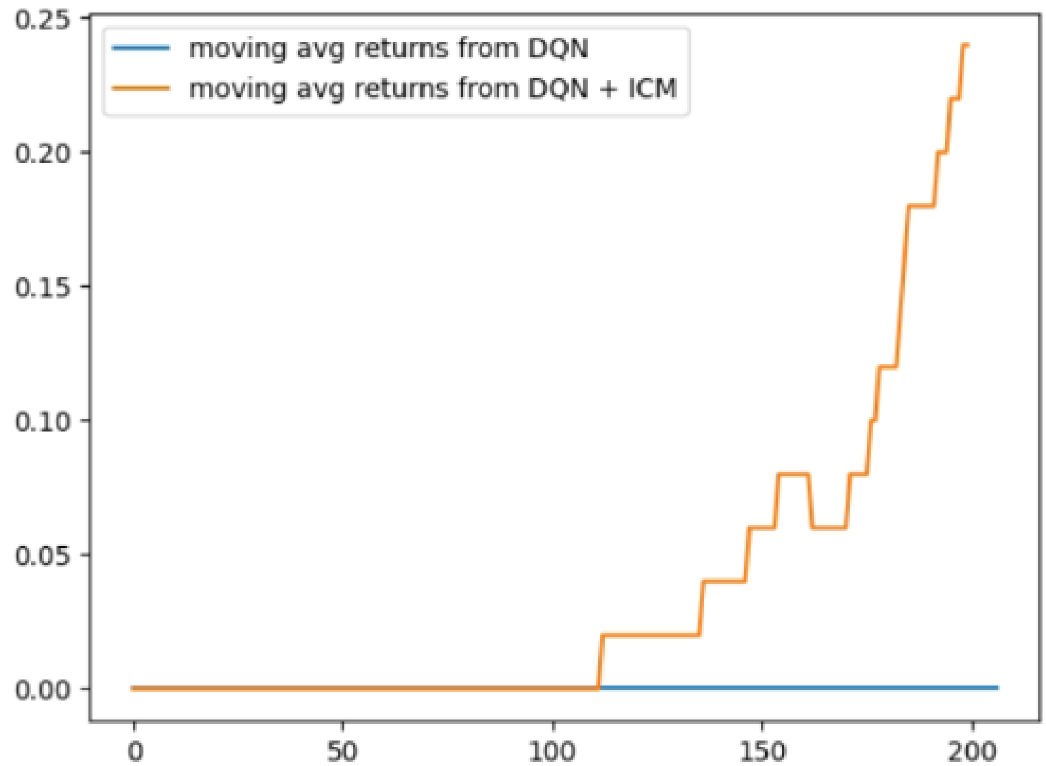

Figure 3: Moving average returns of standalone DQN vs DQN with ICM: FrozenLake8x8.

5.2 Diffusion Augmentation

With goal-oriented expert data, diffusion model consistently generates goal-directed trajectories, resulting in high success rate and efficient paths.

- Data efficiency: Only a small number of near-goal transitions are required; the majority of training comes from ICM-driven exploration.
- Scalability: The approach naturally extends to longer-horizon environments, as diffusion models can generalize from arbitrary start and end windows.
- : Optimality: Sampling multiple trajectories per start-goal pair allows the agent to explore alternative paths, allowing us to choose the best one based-off our criterion (distance from goal).

6 Conclusion

The proposed framework provides a data-efficient and generalizable solution to sparse-reward challenges, with minimal expert data, demonstrating that the combination of local exploration (ICM) and generative planning (diffusion) can extend the capabilities of standard value-based RL. Future work can explore applying this framework to partially observable, continuous-action environments with larger state spaces where long-horizon planning and sparse rewards are even more pronounced.

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